MILITARY SPECIFICATION

CELLULOSE, KRAFT PROCESS WOODPULP

This specification is approved for use by the US Army Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification establishes the requirements and quality assurance provisions for the manufacture and acceptance of one grade of Kraft process cellulose woodpulp for use in combustible cartridge cases.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. - Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1315

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2.2 Other publications. - The following documents form a part of this specification to the extent specified herein.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI) Test Methods

T-233 hm-82	Fiber Length of Pulp by Classification
T-230 om-82	Viscosity of Pulp
T-429 os-78	Alpha-cellulose in Paper
T-223 os-78	Pentosans in Wood and Pulp
T-236 os-76	Kappa Number of Pulp
T-211 om-80	Ash in Wood and Pulp

(Application for copies should be made to The Technical Association of the Pulp and Paper Industry, Technology Park, P.O. Box 105113, Atlanta, GA, 30348).

2.3 Order of precedence. - In the event of conflict between the text of this specification and the referenced specifications cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. - The kraft process cellulose woodpulp shall comply with requirements given in this detailed specification as well as applicable drawings and referenced specifications.

3.2 Material. - The material shall be kraft process cellulose woodpulp, meeting the requirements of this specification.

3.3 Properties. - The properties of the kraft process cellulose woodpulp shall be as specified in Table I when tested in accordance with the applicable test methods:

TABLE I

Property	Minimum	Maximum	Applicable Test Method
Fiber Length (mm)	2	3.5	4.3.1
Viscosity (centipiose)	10	25	4.3.2
Alpha Cellulose (%)	85		4.3.3
Pentosans (%)	5		4.3.4
Lignin (%)		0.5	4.3.5
Ash Content		0.25	4.3.6

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3.4 Workmanship. - The best commercial practices shall be used in the manufacture of the product furnished under this specification. Procedures and controls shall be employed which preclude conditions which may adversely affect the safety and reliability of the product when tested in accordance with 4.3.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such specifications are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.2 Quality Conformance Inspection

4.2.1 Lot formation. - A lot shall consist of one or more batches of kraft process cellulose woodpulp produced by one manufacturer, in accordance with the same specification, or same specification revision, under one set of operating conditions.

4.2.2 Testing. - It is emphasized that all applicable safety rules, regulations, practices and procedures be followed in handling and processing the kraft process cellulose woodpulp.

4.2.2.1 Sampling. - Sampling shall be in accordance with MIL-STD-105.

4.3 Test methods and procedures. - All tests given in this section shall be performed using prescribed procedures given in the Technical Association of the Pulp and Paper Industry (TAPPI) Official Test Methods. See 6.3 for use of equivalent test methods.

4.3.1 Fiber length. - The sample shall be tested for fiber length in accordance with TAPPI Test method T-233 hm-82.

4.3.2 Viscosity. - The sample shall be tested for viscosity in accordance with TAPPI method TM-230 om-82.

4.3.3 Alpha cellulose. - The sample shall be tested for Alpha Cellulose in accordance with TAPPI Test method T-429 os-78.

4.3.4 Pentosans. - The sample shall be tested for pentosans in accordance with TAPPI Test method T-223 os-78.

4.3.5 Lignin. - The sample shall be tested for lignin in accordance with TAPPI Test method T-236 os-76.

4.3.6 Ash content. - The sample shall be tested for Ash Content in accordance with TAPPI Test method T-211 om-80.

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4.3.7 Workmanship. - The sample, when visually examined, shall be free from dirt, chips and foreign matter.

5. PACKAGING

5.1 Packing. - The kraft woodpulp shall be packed in accordance with standard commercial practices.

5.2 Marking. - Unless otherwise specified, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Kraft Process Cellulose Woodpulp is intended to be used in combustible cartridge cases and other combustible ordnance products.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

6.3 Equivalent test methods. - The test methods given in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent, (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Technical Agency to: COMMANDER, ARDC, ATTN: AMSMC-QAR-R, Dover, NJ 07801. This description should include, but not be limited to, the procedures used, the accuracy and precision, and drawings of any special equipment required.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-A774)

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